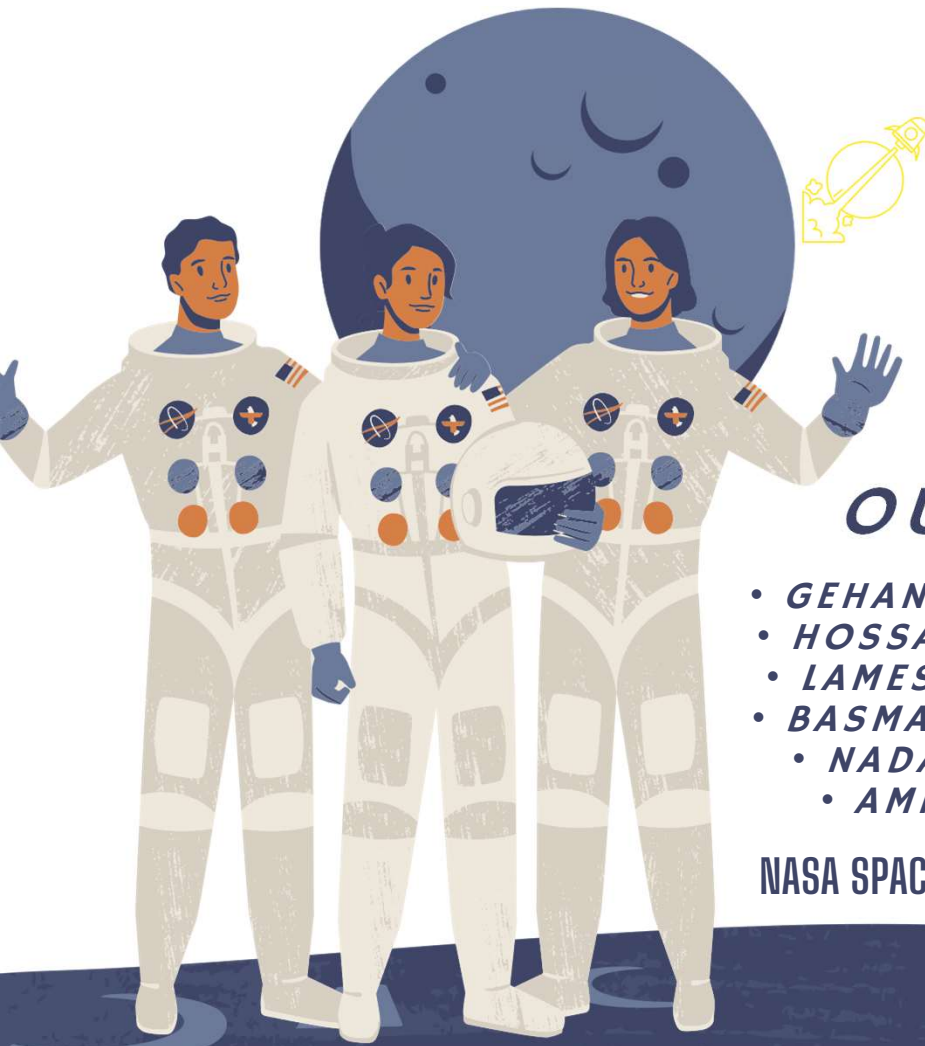


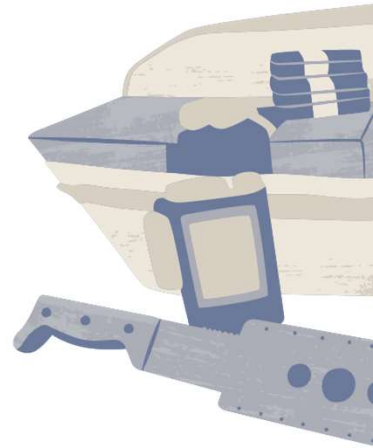


CUREX

OUR TEAM:

- *GEHAN MAHMOUD*
- *HOSSAM HASSAN*
- *LAMES MOSTAFA*
- *BASMALLA AHMED*
 - *NADA WALEED*
 - *AMR TAMER*

NASA SPACE APPS 2025.



PROBLEM STATEMENT

The harsh conditions in space, such as complete isolation, physical and psychological exhaustion, and lack of gravity, affect the health of astronauts. With the lack of medical tools in space, this makes the matter more difficult for astronauts and affects their ability to perform their missions well. Astronauts also rely on taking painkillers continuously, which in the long run affect the kidneys and other parts of the body.

Cases like this seem to have similar symptoms to people suffering from rheumatoid arthritis. These conditions can cause joint pain, muscle stiffness, and inflammation due to prolonged exposure to zero gravity, which weakens bones and muscles.

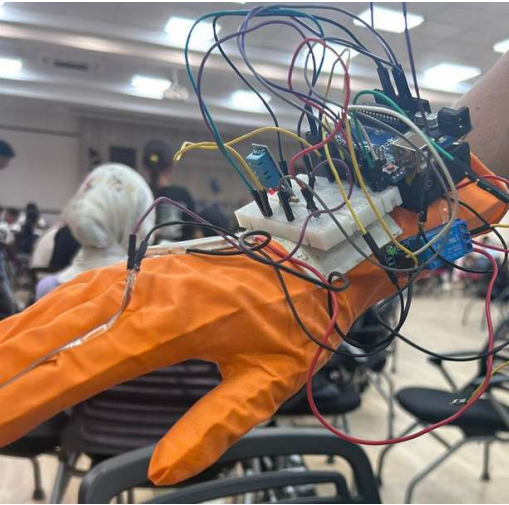
With the absence of advanced medical equipment, astronauts have limited treatment options.

Constant use of medications causes side effects and reduces their efficiency during missions.

Therefore, there is an increasing need for a smart and safe solution that relieves and maintains pain without medications.



SOLVE



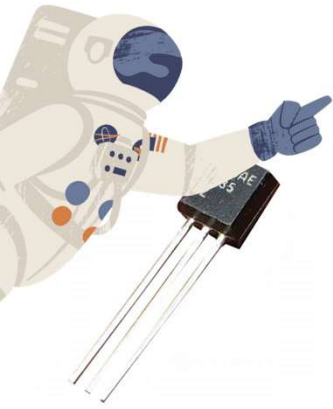
Smart gloves help relieve pain without resorting to medications that have side effects. Gloves relieve pain. If there is muscle stiffness or chronic pain (such as rheumatoid arthritis or tension), → heat helps:

- Stimulating blood circulation.
- Expansion of blood vessels.

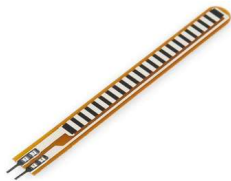
Reducing muscle spasms. In better space we use “Hot-Cold Therapy”: Its main function (cooling): It reduces blood flow to the affected area → this reduces inflammation and swelling. It reduces the activity of sensory nerves → so I feel less pain. It prevents the worsening of swelling resulting from pressure or effort. It reduces inflammation and swelling, which helps astronauts perform their tasks safely and helps them be in habitats safely. Real-time monitoring through sensors that measure temperature, swelling, etc., and displays real-time data on the mobile phone. Cooling from 15°C to 20°C reduces inflammation and swelling. Heating from 38°C to 42°C. It reactivates blood circulation and prevents muscle stiffness



Technology & Components



LM35DZ



**FLEX
SENSOR**



**ARDUINO
UNO**



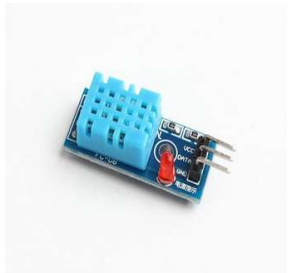
PELTIER



WIRES



GLOVES



DHT11 SENSOR



ESP32



BATTERY



HOLDER



BREADBOARD

- DHT11 → Sensor (Temperature & Humidity).
- LM35 → Sensor (Body temperature).
- FSR & Flex → Sensors (Swelling & Finger bending).
- WI-FI → Communication (Real-time data).
- Peltier → Actuator (Cooling & Heating for pain relief)
 - As micro controller

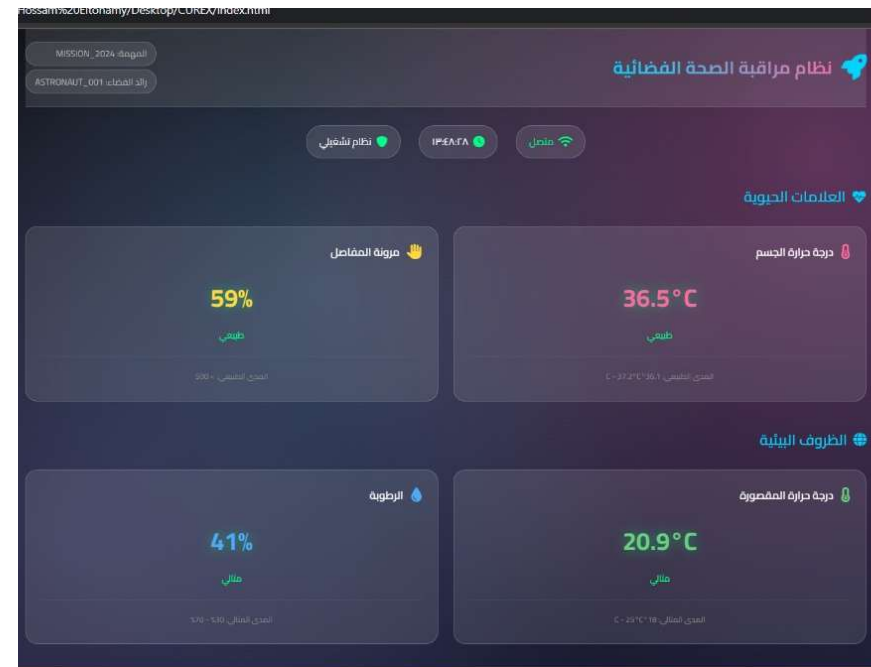
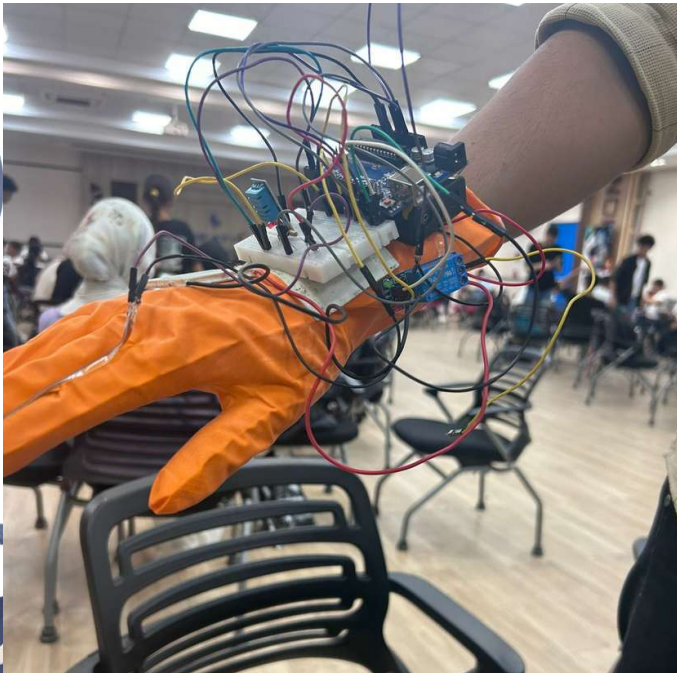




HOW IT WORKS:

- Sensors → Data collection.
- WiFi → send data.
- Peltier → Heating & cooling therapy.
- App → Display health status

prototype & app



BENEFITS & IMPACTS



- Less drug dependency → safer missions.
- Healthier astronauts → higher efficiency.
- Real-time monitoring → sustainable space health

In space: It helps astronauts complete their missions without suffering from pain or the side effects of painkillers .

On the earth: It is used to relieve pain in rheumatoid patients and the elderly It helps astronauts perform their tasks without suffering from pain or the positive effects of painkillers.

It enables you to rely on more medications → making things safer and more sustainable in the long term.

Maintaining astronauts' health and mental focus → which increases their production and work capabilities. It helps astronauts perform their tasks without suffering from pain or the positive effects of painkillers.

It enables you to rely on more medications → making things safer and more sustainable in the long term.

Maintaining astronauts' health and mental focus → which increases their production and work capabilities.

Future Work.

Adding a MAX30102 sensor to measure the pulse and oxygen levels in the blood and connecting it with the rest of the sensors on the phone application for periodic monitoring of the general health of the astronauts. Implementing the project in a glove suit is just an experimental version, but adding the project in the suit where it has joints such as the knee, elbow and shoulder joints to relieve pain in general, adding artificial intelligence algorithms to analyze sensor data (temperature, pulse, swelling...) and predict the occurrence of pain or inflammation before they occur, so that the system automatically intervenes with appropriate heat or cooling.

